

HOW TO GET MASSES FROM EXTRA DIMENSIONS

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A generalized method of dimensional reduction, applicable to theories in curved space, is described. As in previous works by other authors, the extra dimensions are related to the manifold of a Lie group. The new feature of this work is to define and study a class of Lie groups, called “flat groups”, for which the resulting theory has no cosmological constant, a well-behaved potential, and a number of arbitrary mass parameters. In particular, when the analysis is applied to the reduction of 11-dimensional supergravity to four dimensions it becomes possible to incorporate three arbitrary mass parameters in the resulting $N = 8$ theory. This shows that extended supersymmetry theories allow more possibilities for spontaneous symmetry breaking than was previously believed to be the case.

1. Introduction

Dimensional reduction has been used to construct Yang-Mills theories [1,2] with extended supersymmetry [3] and the $N = 8$ supergravity theory [4]. In “ordinary” dimensional reduction, the coordinates of a $D + E$ dimensional theory are divided into D space-time coordinates (x^μ) and E internal coordinates (y^α) that form a compact space. The fields and symmetry transformation laws that the theory possesses in $D + E$ dimensions are taken to be y independent. This requirement for the fields and transformation laws implies in particular that an invariance of the $D + E$ dimensional theory results in a corresponding one for the reduced theory in D dimensions. For example, from a theory with simple supersymmetry in $D + E$ dimensions one deduces a theory with extended supersymmetry in D dimensions ***. An obvious limitation of this approach is that starting from a massless theory (as is necessarily the case for supersymmetric theories in 10 or 11 dimensions [3,6]) the resulting reduced theory is also massless (aside from possible masses due to the Higgs mecha-

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*** For a review of extended supergravity theories, see ref. [5].

nism or topological excitations). Also it is exactly invariant under the extended supersymmetry algebra, implying Fermi-Bose degeneracy that needs to be broken to describe the real world.

In a recent paper [7], a generalization of the dimensional reduction method was formulated that makes it possible to obtain reduced theories having mass parameters and spontaneously broken symmetries. The new ingredient is to allow the fields and transformation laws to depend upon the internal y coordinates in a well-defined fashion determined by a symmetry in $D + E$ dimensions. The y dependence must be chosen such that the fields define fibre bundles that are continuous on the E -dimensional compact space. When the space has non-trivial topology, the fields viewed as functions of y need only be continuous on coordinate patches. It then turns out that the y dependence essentially cancels in the Lagrangian and can be factored out of the transformation laws. The important consequence of this procedure is that the y -dependence of the fields results in mass terms and new couplings in D dimensions.

Two types of symmetries that can be exploited in a generalized dimensional reduction were distinguished in ref. [7]: “external” symmetries having nothing to do with space-time in $D + E$ dimensions and “internal” symmetries that do involve space-time. Only external symmetries were explicitly used: a chiral $U(1)$ symmetry in the reduction of massless $N = 1$ supergravity theory in four dimensions to yield a spontaneously broken $N = 2$ supergravity theory in three dimensions and the non-Abelian gauge invariance of the Yang-Mills action for which the generalized dimensional reduction technique gave a reinterpretation of the usual Higgs mechanism. In the physically more interesting cases of supergravity in 10 or 11 dimensions there is no external symmetry that can be used in the dimensional reduction to generate masses. In this paper it is shown how to use the “internal” invariance under general coordinate transformations of the $D + E$ dimensional theory to introduce a non-trivial dependence upon the y coordinates that gives rise to masses and spontaneous supersymmetry breaking. The basic idea is to relate the y ’s to coordinates of an E -dimensional Lie group [8–10]. The general coordinate transformations are then restricted to be general x -dependent coordinate transformations in D -dimensional space-time plus x - and y -dependent coordinate transformations on the group manifold. A corresponding restriction is introduced for the fields in such a way that the y dependence cancels in the action (aside from a necessary density function) and factors out of the transformation laws of the fields. In the case of pure gravity the resulting action in D dimensions is identical to one obtained previously by Cho and Freund [10].

In general, the reduced gravity theory contains a cosmological constant or, worse yet, a potential depending on scalar fields that is unbounded from below. There exists a class of groups, called “flat groups”, for which neither problem arises. In the case of flat groups some of the scalar fields are “eaten” by vectors that become massive, and most of the remaining scalars also become massive at the same time. The minimum value of the potential is exactly zero, so the theory contains no cosmological constant or tachyons. For a particular type of flat group, the reduction from

eleven to four dimensions allows three arbitrary mass parameters to be introduced.

The application of generalized dimensional reduction to various matter fields coupled to gravity is also studied. Those kinds which are present in ten- or eleven-dimensional supergravity [6] are given special attention. In spite of various dangers that threaten, the reduced action has the necessary gauge invariances of a consistent theory in each case. One feature of the theories derived is that their pattern of masses is entirely determined by the group G and thus, in a sense, is purely geometrical.

The paper is organized as follows. In sect. 2 the ordinary and generalized methods of dimensional reduction are described for pure gravity. Flat groups are introduced and the mass patterns that result are analyzed. Sect. 3 explains the reduction when various kinds of matter fields are coupled to gravity. In particular, the reduction from 11 to 4 dimensions of a Majorana spin- $\frac{3}{2}$ field is discussed, and it is shown that the reduction can give mass to each of the 8 gravitinos of the $N = 8$ theory. Finally, sect. 4 summarizes the results and discusses various remaining problems. While many of the ingredients required for the construction of spontaneously broken $N = 4$ and $N = 8$ supergravity are given, a detailed analysis of those theories is left for the future.

2. Dimensional reduction of Einstein's theory of gravitation

2.1. Notation and conventions

The starting point is the standard Einstein action in $D + E$ dimensions. The *viel-bein* formalism is used, as the main interest is in the application of these results to supergravity theory. The notation is the following: for general indices running over $D + E$ directions of space-time "hatted" symbols are used: Greek hatted letters for curved (base space) indices ($\hat{\mu}, \hat{\nu}, \hat{\rho}, \dots$) and Latin hatted letters for flat (tangent space) indices ($\hat{m}, \hat{n}, \hat{p}, \dots$). For space-time indices taking D values, late Greek letters are used in the curved case ($\mu, \nu, \rho, \dots$) and late Latin letters in the flat case ($m, n, p, \dots$). For internal indices taking E values, early Greek letters are used in the curved case ($\alpha, \beta, \gamma, \dots$) and early Latin letters in the flat case ($a, b, c, \dots$). The general coordinates $x^{\hat{\mu}}$ thus consist of space-time coordinates x^{μ} and internal coordinates y^{α} . The flat Lorentz metric of the tangent space is chosen to be $(+ - \dots -)$ with the internal dimensions all space-like. They are assumed to form a compact space.

The Einstein action is given by

$$S = -\frac{1}{4\kappa^2} \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} \hat{R}(\hat{\omega}). \quad (1)$$

The use of the hat symbol designates a $D + E$ dimensional object. Without it, one is

referring to a D -dimensional object ^{*}. $\hat{R}$ is the usual Einstein curvature scalar constructed from the inverse *vielbein* and the curvature tensor

$$\hat{R} = \hat{V}^{\hat{\mu}\hat{\nu}} \hat{V}^{\hat{\rho}\hat{s}} \hat{R}_{\hat{\mu}\hat{\nu}\hat{\rho}\hat{s}}, \quad (2)$$

where

$$\hat{R}_{\hat{\mu}\hat{\nu}\hat{\rho}\hat{s}} = \partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{\rho}\hat{s}} + \hat{\omega}_{\hat{\mu}\hat{\rho}}^{\hat{t}} \hat{\omega}_{\hat{\nu}\hat{s}\hat{t}} - (\hat{\mu} \leftrightarrow \hat{\nu}). \quad (3)$$

$\hat{V}$ represents the determinant of the *vielbein* $\hat{V}_{\hat{\mu}}^{\hat{\nu}}$. In the normalization of (1) we have introduced $\rho(E)$, the invariant volume of the internal space. It has the dimensions of $(\text{length})^E$, so that $\kappa^2/4\pi$ is the usual Newton's constant in D dimensions having dimensions of $(\text{length})^{D-2}$. The Lagrangian density $\hat{V}\hat{R}$ will be arranged to become y independent, except for a measure $U(y)$ to be described later, so that $\rho(E)$ can be conveniently chosen such that

$$\rho(E) = \int U(y) d^E y. \quad (4)$$

It is convenient to work with the Latin connection $\hat{\omega}_{\hat{t}\hat{r}\hat{s}} = \hat{V}_{\hat{t}}^{\hat{\nu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}}$, which is a scalar under general coordinate transformations and is antisymmetric in its last two indices. In the second-order formalism (which we use throughout) one has

$$\hat{\omega}_{\hat{t}\hat{r}\hat{s}} = -\hat{\Omega}_{\hat{t}\hat{r},\hat{s}} + \hat{\Omega}_{\hat{r}\hat{s},\hat{t}} - \hat{\Omega}_{\hat{s}\hat{t},\hat{r}}, \quad (5)$$

where

$$\hat{\Omega}_{\hat{r}\hat{s},\hat{t}} = \frac{1}{2} (\hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{s}}^{\hat{\nu}} - \hat{V}_{\hat{s}}^{\hat{\mu}} \hat{V}_{\hat{r}}^{\hat{\nu}}) \partial_{\hat{t}} \hat{V}_{\hat{\mu}\hat{\nu}}. \quad (6)$$

A formula for Einstein's action in terms of the Latin connection simplifies the task of dimensional reduction appreciably. Integrating by parts the $\partial\omega$ terms, Einstein's action becomes ^{**}

$$S = -\frac{1}{4\kappa^2} \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} (\hat{\omega}_{\hat{t}\hat{r}}^{\hat{s}} \hat{\omega}^{\hat{s}\hat{t}}_{\hat{r}} + \hat{\omega}_{\hat{r}\hat{t}}^{\hat{s}} \hat{\omega}^{\hat{s}\hat{t}}_{\hat{s}}). \quad (7)$$

Since each term in (7) is a world scalar (there are no Greek indices) the general coordinate invariance is manifest. The integration by parts on the y variables could in principle introduce a surface-term correction to (7). However, it turns out that the requirement of general coordinate invariance restricts the form of the y -dependence in the general reduction to one for which the surface terms vanish. This will be explained in subsect. 2.3. First, the ordinary dimensional reduction of gravity without y dependence is reviewed. This allows a sharper comparison with the results obtained from the generalized reduction.

^{*} The notation differs from that of ref. [7] where capitalization played the role that the hat plays here.

^{**} This formula, valid in the second-order formalism, would have to be corrected by torsion terms in the first-order formalism.

2.2. Ordinary dimensional reduction

In the “ordinary” dimensional reduction all fields are taken to be y independent ^{*}. Here the fields are just the various components of the *vielbein* and so $\hat{V}_\mu^\tau(x, y) = \hat{V}_\mu^\tau(x)$. In other words derivatives ∂_α vanish. Furthermore because of the local Lorentz invariance in $D + E$ dimensions it is possible to choose a triangular parametrization of the *vielbein* having the form

$$\hat{V}_\mu^\tau = \begin{pmatrix} \delta^\gamma V_\mu^\tau & 2\kappa A_\mu^\alpha \Phi_\alpha^\tau \\ 0 & \Phi_\alpha^\tau \end{pmatrix}. \quad (8)$$

In this formula $\delta = \det \Phi$ and the power γ is a free parameter, corresponding to an arbitrary Weyl transformation of the D -dimensional *vielbein*, to be chosen later. The *vielbein* in eq. (8) contains a D -dimensional *vielbein* (graviton), E vector fields A_μ^α and E^2 scalar fields Φ_α^τ . However, due to local Lorentz invariance in the internal directions, only $\frac{1}{2}E(E + 1)$ of the scalar fields correspond to propagating particles.

The $D + E$ dimensional theory is invariant under the general coordinate transformations

$$\delta \hat{V}_\mu^\tau = \xi^{\hat{\rho}} \partial_{\hat{\rho}} \hat{V}_\mu^\tau + \partial_{\hat{\mu}} \xi^{\hat{\rho}} \hat{V}_{\hat{\rho}}^\tau. \quad (9)$$

Corresponding to the assumed y independence of the fields it is appropriate to take the infinitesimal parameters $\xi^{\hat{\mu}}$ also to be y independent. Then under $\xi^{\hat{\mu}}(x)$ transformations Φ_α^τ transforms as a scalar, while A_μ^α and V_μ^τ transform as vectors. For the $\xi^\alpha(x)$ transformations one finds

$$\delta V_\mu^\tau = \delta \Phi_\alpha^\tau = 0, \quad (10a)$$

$$\delta A_\mu^\alpha = \frac{1}{2\kappa} \partial_\mu \xi^\alpha. \quad (10b)$$

Thus the D -dimensional relic of the $\xi^\alpha(x, y)$ invariances, restricted to y -independent transformations, is a set of E Abelian gauge invariances. In order to prove that the theory has D -dimensional general coordinate invariance and this $[U(1)]^E$ invariance, one observes that

$$\delta \int d^D x \int d^E y \hat{V} \hat{R} = \int d^D x \int d^E y \partial_{\hat{\mu}} (\xi^{\hat{\mu}} \hat{V} \hat{R}). \quad (11)$$

The vanishing of this total divergence is guaranteed for the D space-time indices because this integration has no surface term and for the E internal indices by the y independence of $\xi^{\hat{\mu}}$ and the other factors.

Having chosen the y dependence of the fields and the transformation laws, the dimensional reduction proceeds by computing the various quantities appearing in

^{*} As explained in refs. [2,7], this language summarizes the consequences of a normal mode expansion on the compact space and the dropping of all fields that become infinitely massive in the limit that the size of the compact space is shrunk to zero.

the action: inverse *vielbein*, metric (which is useful for the coupling to other fields), inverse metric, and from them the various possible connections. One advantage of using the flat connections is that the $[U(1)]^E$ gauge invariance is manifest. The inverse *vielbein*, defined by

$$\hat{V}_r^{\hat{\mu}} \hat{V}_{\hat{\mu}\hat{s}} = \eta_{\hat{r}\hat{s}} , \quad (12)$$

is found to be

$$\hat{V}_{\hat{r}}^{\hat{\mu}} = \begin{pmatrix} \delta^{-\gamma} V_r^{\mu} & -2\kappa \delta^{-\gamma} A_r^{\alpha} \\ 0 & \Phi_a^{\alpha} \end{pmatrix}, \quad (13)$$

where Φ_a^{α} is the inverse internal *vielbein*. The metric and its inverse are given by

$$\hat{g}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} \delta^{2\gamma} g_{\mu\nu} + 4\kappa^2 A_{\mu}^{\gamma} A_{\nu}^{\gamma} & -2\kappa A_{\mu\beta} \\ -2\kappa A_{\nu\alpha} & -h_{\alpha\beta} \end{pmatrix}, \quad (14a)$$

$$\hat{g}^{\hat{\mu}\hat{\nu}} = \begin{pmatrix} \delta^{-2\gamma} g^{\mu\nu} & -2\kappa A^{\mu\beta} \\ -2\kappa A^{\nu\alpha} & -h^{\alpha\beta} + 4\kappa^2 \delta^{-2\gamma} A^{\rho\alpha} A_{\rho}^{\beta} \end{pmatrix}, \quad (14b)$$

where Greek internal indices are raised and lowered by the “internal metric” field

$$h_{\alpha\beta} = \Phi_{\alpha}^a \delta_{ab} \Phi_{\beta}^b . \quad (15)$$

$h_{\alpha\beta}$ is a positive-definite metric, and to account for this choice a minus sign has been supplied for each h . The determinant of the *vielbein* is given by

$$\hat{V} = \delta^{\gamma D+1} V , \quad (16)$$

where $V = \det V_{\mu}^{\tau}$. There are six Latin connection coefficients to calculate. One obtains

$$\hat{\omega}_{p,mn} = \delta^{-\gamma} [\omega_{p,mn} + \gamma(\eta_{pm} V_n^{\lambda} - \eta_{pn} V_m^{\lambda}) \partial_{\lambda} \ln \delta] , \quad (17a)$$

$$\hat{\omega}_{p,ma} = \kappa \delta^{-2\gamma} F_{pm}^{\alpha} \Phi_{\alpha a} , \quad (17b)$$

$$\hat{\omega}_{p,ab} = \frac{1}{2} \delta^{-\gamma\gamma} \Phi_a^{\alpha} V_p^{\mu} \partial_{\mu} \Phi_{\alpha b} - (a \leftrightarrow b) , \quad (17c)$$

$$\hat{\omega}_{c,mn} = -\kappa \delta^{-2\gamma} F_{mn}^{\alpha} \Phi_{\alpha c} , \quad (17d)$$

$$\hat{\omega}_{c,ma} = \frac{1}{2} \delta^{-\gamma} \Phi_a^{\alpha} \Phi_c^{\beta} V_m^{\lambda} \partial_{\lambda} h_{\alpha\beta} , \quad (17e)$$

$$\hat{\omega}_{c,ab} = 0 , \quad (17f)$$

where

$$F_{\mu\nu}^{\alpha} = \partial_{\mu} A_{\nu}^{\alpha} - \partial_{\nu} A_{\mu}^{\alpha} . \quad (18)$$

The gauge invariance of these expressions is obvious.

Expanding out $\hat{R}$ we see that the coefficient of R contains δ raised to the power

$\gamma(D-2)+1$. Thus, by choosing

$$\gamma = -\frac{1}{D-2}, \quad (19)$$

one brings the Einstein term of the reduced theory to the usual form. Altogether, substituting eqs. (17) and (19) into eq. (7) and dropping the y integration one obtains

$$\begin{aligned} S = \int d^D x \, V \left\{ -\frac{1}{4\kappa^2} R - \frac{1}{4} \delta^{2/(D-2)} F^{\mu\nu\alpha} F_{\mu\nu}^\beta h_{\alpha\beta} \right. \\ \left. - \frac{1}{16\kappa^2} g^{\rho\lambda} \partial_\rho h_{\alpha\beta} \partial_\lambda h^{\alpha\beta} \right. \\ \left. + \frac{1}{4\kappa^2(D-2)} g^{\rho\lambda} \partial_\rho \ln \delta \partial_\lambda \ln \delta \right\}. \end{aligned} \quad (20)$$

The reduced theory contains only massless particles: a graviton, E Abelian gauge fields (whose ghosts are decoupled because of the $[U(1)]^E$ gauge invariance), and $\frac{1}{2}E(E+1)$ scalars. Expanding $h_{\alpha\beta}$ about $\delta_{\alpha\beta}$ (its vacuum value), one verifies easily that the vector and scalar kinetic terms have the correct sign to describe particles of positive metric. The local invariances of the action are general coordinate invariance in D dimensions, local Lorentz invariance (which is trivial until fermions are added), and local $[U(1)]^E$ gauge invariance. There is also a global $SL(E, R)$ invariance since the action is invariant under the finite transformations

$$h_{\alpha\beta} \rightarrow T_{\alpha}^{\alpha'} h_{\alpha'\beta'} T_{\beta'}^{\beta}, \quad (21a)$$

$$A_{\mu}^{\alpha} \rightarrow (T^{-1})_{\alpha}^{\alpha'} A_{\mu}^{\alpha'}, \quad (21b)$$

where T is a real unimodular $E \times E$ matrix.

2.3. Generalized dimensional reduction

The analysis of subsect. 2.2 can be generalized by allowing the fields to depend upon the internal y coordinates. This dependence must satisfy a number of criteria. First, it should be possible to define a limit in which the fields become y independent and the results of subsect. 2.2 are recovered. This implies that the number of physical modes is unchanged, which is especially important in the case of supergravity where we know that the counting of degrees of freedom must be the same as that obtained in subsect. 2.2. Second, the y -dependence of the fields and transformation laws must be of a simple factorizable form that can be factored out of the transformation laws. Only a very simple y dependence (to be described below) may survive in the action. Third, the reduced theory should contain no ghost particles. In subsect. 2.4 we will explore the additional requirements of a stable vacuum and no cosmological term.

The second criterion implies that the y dependence cancels because of a symmetry. Since local Lorentz invariance is quite irrelevant in the absence of fermions, the only available invariance of Einstein's action to exploit is general coordinate invariance. The generalized reduction is uniquely specified by choosing the y dependence of the infinitesimal parameters $\xi^{\hat{\mu}}(x, y)$. The choice should be made in such a way as to describe the general coordinate transformations in D -dimensional space-time and local gauge transformations such that the $A_{\mu}^{\alpha}(x)$ are gauge fields for as general a Lie group as possible. The algebra of general coordinate transformations is $[\delta_{\xi_1}, \delta_{\xi_2}] = \delta_{\xi_3}$, where

$$\xi_3^{\hat{\mu}}(x, y) = \xi_2^{\hat{\mu}}(x, y) \partial_{\hat{\rho}} \xi_1^{\hat{\rho}}(x, y) - (1 \leftrightarrow 2). \quad (22)$$

If we now set

$$\xi^{\mu}(x, y) = \xi^{\mu}(x), \quad (23a)$$

$$\xi^{\alpha}(x, y) = [U^{-1}(y)]_{\beta}^{\alpha} \xi^{\beta}(x), \quad (23b)$$

then we recover the results of subsect. 2.2 in the limit that the non-singular $E \times E$ matrix U approaches unity. The commutator of two space-time general coordinate transformations has the usual algebra given by the x -space analog of eq. (22). Commuting a space-time transformation with parameter $\xi_1^{\alpha}(x)$ and an internal transformation with parameter $\xi_2^{\alpha}(x)$ gives a new internal transformation with parameter

$$\xi_3^{\alpha}(x) = -\xi_1^{\rho}(x) \partial_{\rho} \xi_2^{\alpha}(x), \quad (24)$$

which just means that the parameters of an internal transformation are space-time scalars. Finally, and most importantly, the commutator of two internal transformations is given by

$$\xi_3^{\gamma}(x) = f^{\gamma}_{\alpha\beta} \xi_1^{\alpha}(x) \xi_2^{\beta}(x), \quad (25)$$

where

$$f^{\gamma}_{\alpha\beta} = (U^{-1})_{\alpha}^{\alpha'} (U^{-1})_{\beta}^{\beta'} (\partial_{\beta'} U_{\alpha'}^{\gamma} - \partial_{\alpha'} U_{\beta'}^{\gamma}). \quad (26)$$

The essential point, of course, is that the U 's must have the property that the y dependence cancels in eq. (26) to give f 's that are constants.

It is in fact very easy to define suitable U matrices. By considering the y 's to be a system of coordinates on the manifold of a Lie group G having E generators, one can always represent the infinitesimal generators of the group in terms of differential operators of the form *

$$L_{\alpha}(y) = (U^{-1})_{\alpha}^{\beta} \partial_{\beta}. \quad (27)$$

* This does *not* mean that the manifold of the internal y coordinates is the group manifold. We are mostly interested in non-compact groups defined by compact y manifolds. We are indebted to D. Olive for a discussion of this point.

Since the action of a Lie group on itself constitutes a non-singular change of basis it follows that the U matrix is non-singular. Furthermore, Lie's second theorem [11] states that if

$$[L_\alpha, L_\beta] = f^\gamma_{\alpha\beta} L_\gamma, \quad (28)$$

then the structure constants f are really y independent and given by eq. (26). So given any E -dimensional Lie group G , one can always find U 's such that the f 's are constants. Explicit computation of the U 's is not necessary since the formulas in the dimensional reduction will only involve the structure constants and not the U 's themselves. It will be shown later that restrictions must be placed on the choice of the group.

Through the choice of y dependence described above, an arbitrary E -dimensional Lie algebra has been extracted out of the group of general coordinate transformations in $D + E$ dimensions. Corresponding choices for the y dependence of the fields are required in order that the reduced action be invariant under these transformations. The appropriate rule turns out to be that every lower internal Greek index must be accompanied by a $U(y)$ factor and every upper internal Greek index by a $U^{-1}(y)$ factor. Otherwise nothing is changed from the parametrization of the *vielbein* given in eq. (8). The meaning of the symbols is now

$$V'_\mu(x, y) = V'_\mu(x), \quad (29a)$$

$$A^\alpha_\mu(x, y) = [U^{-1}(y)]^\alpha_\beta A^\beta_\mu(x), \quad (29b)$$

$$\Phi^a_\alpha(x, y) = U^\beta_\alpha(y) \Phi^a_\beta(x). \quad (29c)$$

The counting of degrees of freedom has not been changed. Also, whenever two internal Greek indices are contracted without derivatives the y dependence cancels.

One can compute the transformation laws of the various fields by substituting eq. (29) into eq. (9) using eqs. (8) and (26). The y dependence factors out of the equations leaving

$$\delta \Phi^a_\alpha(x) = f^\gamma_{\alpha\beta} \xi^\beta(x) \Phi^a_\gamma(x), \quad (30a)$$

$$\delta \Phi^a_\alpha(x) = f^\alpha_{\beta\gamma} \xi^\beta(x) \Phi^a_\alpha(x), \quad (30b)$$

$$\delta A^\alpha_\mu(x) = \frac{1}{2k} \partial_\mu \xi^\alpha(x) + f^\alpha_{\beta\gamma} \xi^\beta(x) A^\gamma_\mu(x), \quad (30c)$$

$$\delta V'_\mu(x) = 0. \quad (30d)$$

Thus the A^α_μ 's are gauge potentials for the gauge group G with structure constants $f^\alpha_{\beta\gamma}$. The f 's are dimensional parameters (inverse length or mass) and determine both couplings and masses, as will be shown later.

In eq. (8) δ is taken to be the determinant of $\Phi^a_\alpha(x)$ and thus y independent. $\hat{V}$, however, is y dependent

$$\hat{V} = U(y) \delta^{\gamma D+1} V, \quad (31)$$

where

$$U(y) = \det U_\alpha^\beta(y) . \quad (32)$$

It will be shown that $\hat{R}$ is y independent, and thus the only y dependence in the action is the $U(y)$ factor in $\hat{V}$. One can verify that this factor is precisely the invariant measure so that the y integration gives the volume of the manifold assumed to be compact. The factor $\rho(E)$ is taken in eq. (4) to equal the volume so as to cancel against the y integration. One must still verify that the action is invariant under the $\xi^\alpha(x, y)$ transformations. This is essential for decoupling the ghost components of the vector fields. From eqs. (11), (23) and (31) one deduces the constraint

$$\partial_\alpha [(U^{-1})_\beta^\alpha U] = 0 . \quad (33)$$

An easy calculation shows this to be equivalent to the structure constant condition

$$f^\alpha_{\alpha\beta} = 0 . \quad (34)$$

This condition can be satisfied for all semi-simple Lie algebras and for many others as well, but it does exclude some exotic possibilities. When it is satisfied, the action is indeed invariant under the transformations of the local gauge group G . Also, the partial integration used to obtain eq. (7) is then valid.

Now let us compute the Latin connections generalizing the expressions in eq. (17):

$$\hat{\omega}_{p,mn} = \delta^{-\gamma} [\omega_{p,mn} + \gamma(\eta_{pm} V_n^\lambda - \eta_{pn} V_m^\lambda) \partial_\lambda \ln \delta] , \quad (35a)$$

$$\hat{\omega}_{p,ma} = \kappa \delta^{-2\gamma} F_{pm}^\alpha \Phi_{\alpha a} , \quad (35b)$$

$$\hat{\omega}_{p,ab} = \frac{1}{2} \delta^{-\gamma} \Phi_a^\alpha V_p^\mu \mathcal{D}_\mu \Phi_{\alpha b} - (a \leftrightarrow b) , \quad (35c)$$

$$\hat{\omega}_{c,mn} = -\kappa \delta^{-2\gamma} F_{mn}^\alpha \Phi_{\alpha c} , \quad (35d)$$

$$\hat{\omega}_{c,ma} = \frac{1}{2} \delta^{-\gamma} \Phi_a^\alpha \Phi_c^\beta V_m^\lambda \mathcal{D}_\lambda h_{\alpha\beta} , \quad (35e)$$

$$\hat{\omega}_{c,ab} = \frac{1}{2} f^\alpha_{\beta\gamma} (\Phi_a^\beta \Phi_b^\gamma \Phi_{\alpha c} + \Phi_a^\beta \Phi_c^\gamma \Phi_{\alpha b} - \Phi_b^\beta \Phi_c^\gamma \Phi_{\alpha a}) . \quad (35f)$$

In these expressions all the fields are functions of x only, all y dependence having cancelled. Thus $\hat{R}$ is indeed y independent as previously asserted. A new non-zero connection appears in eq. (35f), which will be important for fermion masses in subsect. 3.5, while the others are just non-Abelian generalizations of eq. (17), with the definitions

$$F_{\mu\nu}^\alpha = \partial_\mu A_\nu^\alpha - \partial_\nu A_\mu^\alpha - 2\kappa f^\alpha_{\beta\gamma} A_\mu^\beta A_\nu^\gamma , \quad (36)$$

$$\mathcal{D}_\mu \Phi_\alpha^a = \partial_\mu \Phi_\alpha^a - 2\kappa f^\gamma_{\alpha\beta} A_\mu^\beta \Phi_\gamma^a , \quad (37a)$$

$$\mathcal{D}_\mu \Phi_a^\alpha = \partial_\mu \Phi_a^\alpha - 2\kappa f^\alpha_{\beta\gamma} A_\mu^\beta \Phi_a^\gamma . \quad (37b)$$

$F_{\mu\nu}^\alpha$ is the covariant gauge field strength and $\mathcal{D}_\mu$ is the gauge covariant derivative.

Inserting eqs. (31) and (35) into eq. (1) and using eqs. (4) and (19) one obtains

the reduced action

$$S = \int d^D x \, V \left\{ -\frac{1}{4\kappa^2} R - \frac{1}{4} \delta^{2/(D-2)} F^{\mu\nu\alpha} F_{\mu\nu}^\beta h_{\alpha\beta} \right. \\ \left. - \frac{1}{16\kappa^2} g^{\rho\lambda} \mathcal{D}_\rho h_{\alpha\beta} \mathcal{D}_\lambda h^{\alpha\beta} + \frac{1}{4\kappa^2(D-2)} g^{\rho\lambda} \partial_\rho \ln \delta \partial_\lambda \ln \delta \right. \\ \left. - \frac{1}{16\kappa^2} \delta^{-2/(D-2)} f^\alpha_{\beta\gamma} [2f^\beta_{\alpha\gamma} h^{\gamma\gamma'} + f^{\alpha'}_{\beta'\gamma'} h_{\alpha\alpha'} h^{\beta\beta'} h^{\gamma\gamma'}] \right\}. \quad (38)$$

This formula was obtained by Cho and Freund [10] using the metric formalism. The first four terms are obvious generalizations to the non-Abelian case of eq. (20), while the last term is a potential depending upon the scalar fields $h_{\alpha\beta}$ and the choice of the group G . Additional restrictions required for a well-behaved potential are investigated in subsect. 2.4. Although eq. (38) gives well-defined interactions among gravity, gauge fields and scalar fields, one should not really consider it a unified theory because there is no higher symmetry that transforms the different fields into one another. When Einstein's action is embedded in a higher-dimensional supergravity theory, such as the 11-dimensional theory, there will be such transformations and the reduced action will then constitute a genuinely unified theory in D dimensions.

2.4. Choice of the gauge group G

Aside from a positive coefficient, the potential given by eq. (38) is

$$\mathcal{V}(h) = 2f^\alpha_{\beta\gamma} f^\beta_{\alpha\delta} h^{\gamma\delta} + f^\alpha_{\beta\gamma} f^{\alpha'}_{\beta'\gamma'} h_{\alpha\alpha'} h^{\beta\beta'} h^{\gamma\gamma'}. \quad (39)$$

Whenever G is a semisimple group this potential is unbounded from below*. In the case of a semisimple group the structure constants may be chosen to be completely antisymmetrical. Doing this and considering $h_{\alpha\beta}$ of the special form $h\delta_{\alpha\beta}$, one has $\mathcal{V} \propto h^{-1}$, where the coefficient is a negative constant. $\mathcal{V}$ is still to be multiplied by the factor $\delta^{-2/(D-2)} \propto h^{-E/(D-2)}$, but this doesn't change matters. The overall potential goes to $-\infty$ as $h \rightarrow 0$. It was noted in ref. [10] that by adding a cosmological constant of the appropriate sign to the theory in $D + E$ dimensions the vacuum could be stabilized: that is, the potential then has a well-defined minimum. The minimum value is not zero, however, so the D -dimensional theory defined in this way inherits a cosmological term. In any case this option does not appeal to us because supergravity theories in 10 or 11 dimensions do not have a cosmological term [13].

For the reasons just explained the possibility of a semisimple group must be abandoned. Instead, we insist that the only acceptable groups are "flat groups", which are defined by the requirement that $\mathcal{V}$ be a non-negative function which

* This feature has been encountered previously in the gauging of $SO(4)$ and $SU(4)$ supergravity theories [12].

vanishes for $h_{\alpha\beta} = \delta_{\alpha\beta}$. This means that in addition to eq. (34) the structure constants satisfy the following three equations^{*}:

$$\mathcal{V}(1) = 2f^\alpha_{\beta\gamma}f^\beta_{\alpha\gamma} + f^\alpha_{\beta\gamma}f^\alpha_{\beta\gamma} = 0, \quad (40)$$

$$\frac{\partial \mathcal{V}}{\partial h^{\alpha\beta}}(1) = 2f^{\alpha'}_{\beta'\alpha}f^{\beta'}_{\alpha'\beta} + 2f^{\alpha'}_{\beta'\alpha}f^{\alpha'}_{\beta'\beta} - f^{\alpha}_{\alpha'\beta'}f^{\beta}_{\alpha'\beta'} = 0, \quad (41)$$

$$\mathcal{V}(h) \geq 0. \quad (42)$$

One is not accustomed to Yang-Mills theories based on groups other than products of $U(1)$'s and simple groups, but that is what is required in this problem. The possibility exists because of the non-trivial role played by scalar fields.

Examples of flat groups can be generated by a technique that was suggested in ref. [7]. It was pointed out there that if one first makes an ordinary dimensional reduction from $D + E$ dimensions to $D + 1$ dimensions, then there are external symmetries that can be exploited in a subsequent generalized reduction to D dimensions. Thus consider the theory of eq. (20) for $D + 1$ dimensions (with coordinates x^μ, y^1). It has the global $SL(E - 1, R)$ symmetry described in eq. (21). This symmetry may be exploited in the manner of ref. [7] by setting

$$U^\alpha_\beta(y) = (\exp My^1)_\beta^\alpha, \quad (43)$$

where M is a traceless $E \times E$ matrix with zeros in the first row and column. Substituting this expression in eq. (26) gives

$$f^\gamma_{\alpha\beta} = M^\gamma_\alpha \delta^\beta_1 - M^\gamma_\beta \delta^\alpha_1. \quad (44)$$

In other words, one has an algebra whose generators X have the commutation relations

$$[X_1, X_\alpha] = -M^\beta_\alpha X_\beta, \quad (45a)$$

$$[X_\alpha, X_\beta] = 0, \quad \alpha, \beta \neq 1. \quad (45b)$$

To check whether eq. (44) defines a flat group one substitutes it into eq. (40). This gives the constraint

$$\text{Tr} M(M + M^T) = 0. \quad (46)$$

Therefore M must be chosen to be antisymmetric. This means that only $SO(E - 1)$, the maximal compact subgroup of $SL(E - 1, R)$, may be exploited in the reduction from $D + 1$ to D dimensions. It then turns out that eq. (41) is automatically satisfied, as is easily verified. The positivity condition in eq. (42) is also true, but less trivial to prove. The potential expressed in terms of M is now

$$\mathcal{V}(h) = 2h^{11}\text{Tr}(M^2 - MhMh^{-1}) - 2M^\alpha_\beta M^{\alpha'}_{\beta'} h_{\alpha\alpha'} h^{\beta 1} h^{\beta' 1}. \quad (47)$$

^{*} Actually, eq. (41) is a consequence of the other two, and eq. (40) is a consequence of eq. (41), which is the statement that $R_{\alpha\beta} = 0$ for $A^\alpha_\mu(x) = 0$, $V^\alpha_\mu(x) = \delta^\alpha_\mu$, and $\Phi^\alpha_\mu(x) = \delta^\alpha_\mu$. This explains the use of the word "flat".

Before giving the general proof that $\mathcal{V} \geq 0$, let us consider the simple example $E = 3$. In this case M is necessarily of the form

$$M = m \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}. \quad (48)$$

Inserting this in eq. (47) one finds by explicit calculation that

$$\mathcal{V}(h) = 2m^2 \delta^{-2} [(h_{22} - h_{33})^2 + 4h_{23}^2], \quad (49)$$

which is clearly positive. In this example we can trace the fate of the six scalar fields $h_{\alpha\beta}$. Two are massive, two are eaten by vectors that are massive, and two are massless. The general proof of positivity is easiest in a special gauge. Under ξ^α transformations one has

$$\delta h^{1\gamma} = M_{\alpha\xi}^\gamma \xi^\alpha h^{11} - M_{\beta\xi}^\gamma \xi^1 h^{1\beta}. \quad (50)$$

Therefore since $h^{11} \approx 1$ in the weak field limit, there are constant terms in the variation of some of the $h^{1\gamma}$ fields, the exact number being given by the rank of M . Therefore by going to a suitable basis one may choose a gauge in which this number of $h^{1\gamma}$ fields is set equal to zero (they are eaten by vectors). In this gauge

$$M_{\beta\xi}^\alpha h^{\beta 1} = 0, \quad (51)$$

so that the second term in eq. (47) vanishes. Now consider the symmetric matrix

$$Y_{ab} = \Phi_{\alpha a} M_{\beta}^{\alpha} \Phi_b^{\beta} + (a \leftrightarrow b). \quad (52)$$

Recalling eq. (15), one observes that

$$\text{Tr} Y^2 = 2\text{Tr}(M^2 - MhMh^{-1}) \geq 0, \quad (53)$$

which completes the proof that $\mathcal{V} \geq 0$.

One class of flat groups is given by eq. (44) with M real, antisymmetrical, traceless, and zero in the first row and column. The most general possibility is not known, but a more general class that may be the complete solution (up to an irrelevant change of basis) can be described. Suppose internal Greek indices are divided into two types, barred and unbarred, for the purposes of this paragraph only. A barred index runs over the range $1 \leq \bar{\alpha} \leq N$ and an unbarred index over the range $N+1 \leq \alpha \leq E$. Then consider introducing N matrices, each with $E-N$ rows and columns, $(M_{\bar{\alpha}})_{\beta}^{\gamma}$ such that the generators have the algebra

$$[X_{\bar{\alpha}}, X_{\beta}] = -(M_{\bar{\alpha}})_{\beta}^{\gamma} X_{\gamma}, \quad (54a)$$

$$[X_{\alpha}, X_{\beta}] = [X_{\bar{\alpha}}, X_{\bar{\beta}}] = 0. \quad (54b)$$

The matrices $M_{\bar{\alpha}}$ are taken to be real and antisymmetric. This system is obviously a generalization of eq. (45), which corresponds to $N = 1$. The Jacobi identity requires

in addition that

$$\{M_{\alpha}, M_{\beta}\} = 0. \quad (55)$$

It is then easy to see that G is a flat group.

2.5. Evaluation of masses

As a specific example suppose that E is odd and that the flat group is described by a simple matrix M having rank $E - 1$. (When E is even the maximum possible rank is $E - 2$.) Then as a notational convenience let us restrict Greek indices to the range $2 \leq \alpha \leq E$. By an orthogonal transformation (which represents no restriction on the theory) M can be brought to the canonical form

$$M_{\alpha}^{\beta} = \begin{bmatrix} 0 & m_1 & & & & \\ -m_1 & 0 & & & & \\ & & 0 & m_2 & & \\ & & -m_2 & 0 & \cdots & \\ & \bigcirc & & & 0 & m_{(E-1)/2} \\ & & & & -m_{(E-1)/2} & 0 \end{bmatrix}. \quad (56)$$

The assumption on rank then amounts to requiring that all the m 's are non-zero. It is then convenient to choose the gauge that was used in the positivity proof of subsect. 2.4

$$h^{\alpha 1} = h_{\alpha 1} = 0. \quad (57)$$

The vector $A_{\mu}^1 \equiv A_{\mu}$ will remain massless, but all the other vector fields A_{μ}^{α} acquire mass (by swallowing the corresponding $h^{1\alpha}$). To see the details one must look at the Lagrangian more closely. First using eqs. (36) and (44) one notes that

$$F_{\mu\nu} \equiv F_{\mu\nu}^1 = \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}, \quad (58)$$

$$F_{\mu\nu}^{\alpha} = D_{\mu} A_{\nu}^{\alpha} - D_{\nu} A_{\mu}^{\alpha}, \quad (59)$$

where

$$D_{\mu} A_{\nu}^{\alpha} = \partial_{\mu} A_{\nu}^{\alpha} + 2\kappa A_{\mu} M_{\beta}^{\alpha} A_{\nu}^{\beta} \quad (60)$$

is a covariant derivative for "electrically" charged fields. In the basis described by eq. (56), $A_{\mu}^2 + iA_{\mu}^3$ describes a complex vector field with electric charge $2\kappa m_1$. Similarly $A_{\mu}^4 + iA_{\mu}^5$ has charge $2\kappa m_2$ and so forth.

Now consider the third term in eq. (38). Using the covariant derivatives

$$\mathcal{D}_{\rho} h_{11} = \partial_{\rho} h_{11}, \quad (61a)$$

$$\mathcal{D}_{\rho} h_{\alpha 1} = 2\kappa M_{\gamma}^{\beta} A_{\rho}^{\gamma} h_{\beta \alpha}, \quad (61b)$$

$$\mathcal{D}_{\rho} h_{\alpha\beta} = \partial_{\rho} h_{\alpha\beta} - 2\kappa A_{\rho} (M_{\alpha}^{\gamma} h_{\gamma\beta} + M_{\beta}^{\gamma} h_{\gamma\alpha}) \equiv D_{\rho} h_{\alpha\beta}, \quad (61c)$$

we see that the theory contains the mass term for the vector fields

$$\mathcal{L}_{\text{vector mass}} = \frac{1}{2} V g^{\rho\lambda} h_{\alpha\beta} M_{\gamma}^{\alpha} M_{\delta}^{\beta} A_{\rho}^{\gamma} A_{\lambda}^{\delta} . \quad (62)$$

Therefore $M^T M$ is the (mass)² matrix for the vector fields. $A_{\mu}^2 + iA_{\mu}^3$ has mass m_1 , $A_{\mu}^4 + iA_{\mu}^5$ has mass m_2 , and so forth. In general the mass and charge are related by

$$\mathcal{M} = \frac{1}{2\kappa} |Q| . \quad (63)$$

To study the scalar fields it is convenient to remove the determinant h from the matrix $h_{\alpha\beta}$. So we define the unimodular matrix $\bar{h}_{\alpha\beta}$ by

$$\bar{h}_{\alpha\beta} = h_{\alpha\beta} (h)^{-1/(E-1)} . \quad (64)$$

Then the complete kinetic term for the scalar fields obtained from the third and fourth terms in eq. (38) is

$$\begin{aligned} \mathcal{L}_{\text{scalar kin}} = & \frac{1}{2} V g^{\rho\lambda} \left(\partial_{\rho} \varphi_1 \partial_{\lambda} \varphi_1 + \partial_{\rho} \varphi_2 \partial_{\lambda} \varphi_2 \right. \\ & \left. - \frac{1}{8\kappa^2} D_{\rho} \bar{h}_{\alpha\beta} D_{\lambda} \bar{h}^{\alpha\beta} \right) , \end{aligned} \quad (65)$$

where we have set

$$\varphi_1 = \frac{1}{c_1} \ln h \quad (66a)$$

$$\varphi_2 = \frac{1}{c_2} \ln h h_{11} , \quad (66b)$$

with the constants c_1 and c_2 given by

$$C_1 = \sqrt{8(D-2)} \kappa , \quad (67a)$$

$$C_2 = \sqrt{8(E-1)} \kappa . \quad (67b)$$

The potential term for the scalar fields obtained from the last term in eq. (38) usings eqs. (47), (57), and (66) is

$$\mathcal{L}_{\text{scalar int}} = - \frac{1}{8\kappa^2} V \exp \left(\frac{D-3}{D-2} C_1 \varphi_1 - C_2 \varphi_2 \right) \text{Tr} (M^2 - M \bar{h} M \bar{h}^{-1}) . \quad (68)$$

To see what sort of scalar masses this implies one may set $\bar{h} = \exp(2\kappa S)$ where S is a real symmetric traceless $(E-1) \times (E-1)$ matrix and expand in powers of S . Doing this one discovers that there are $\frac{1}{2}(E-3)$ massless scalars in addition to φ_1 and φ_2 , while all the other scalars are charged and massive if none of the m 's are equal. The results are shown in table 1. The total number of scalar modes is $\frac{1}{2}E(E-1) + 1$, as it should be since $E-1$ modes have been eaten by vectors. Thus applying this reduction to 11-dimensional supergravity, the *elfbein* gives rise in four dimensions to a massless graviton, a massless vector, 4 massless scalars, 3 complex massive vectors, and 9 complex massive scalars.

Table 1
Masses and charges of the scalar fields

Mass	Charge	Number of real scalar fields
0	0	$\frac{1}{2}(E+1)$
$2m_i$	$\pm 4\kappa m_i$	$E-1$
$m_i + m_j, i \neq j$	$\pm 2\kappa(m_i + m_j)$	$\frac{1}{4}(E-1)(E-3)$
$ m_i - m_j , i \neq j$	$\pm 2\kappa(m_i - m_j)$	$\frac{1}{4}(E-1)(E-3)$

3. Reduction with extra matter multiplets

In this section the coupling of matter multiplets to $D + E$ dimensional gravity is considered. In particular, the consequences of the generalized dimensional reduction of the gravity multiplet, described in sect. 2, for the reduction of various types of matter multiplets is examined.

3.1. Scalar field

A scalar field occurs in 10-dimensional supergravity theory, where its kinetic term is of the form

$$S_\phi = \frac{1}{2} \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \phi \partial_{\hat{\nu}} \phi. \quad (69)$$

Since ϕ has no internal Greek indices it must be interpreted as a singlet of the group G , and thus one simply sets $\phi(x, y) = \phi(x)$. Then the reduced action in D dimensions is given by

$$S_\phi = \int d^D x \, V g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi. \quad (70)$$

The power of the scalar field δ has cancelled as a consequence of eq. (19). Also, no mass has been introduced for ϕ by the reduction. In fact we see no way by which such a field could acquire mass, other than through quantum corrections.

3.2. Yang-Mills fields

There are supersymmetric Yang-Mills theories for space-time dimension $D + E \leq 10$ [3]. Although the coupling of these theories to the corresponding supergravity theories has not been explicitly worked out for $D + E > 4$, it certainly can be. In particular, it is possible to couple 10-dimensional supersymmetric Yang-Mills theory to 10-dimensional supergravity. Ordinary dimensional reduction then yields $N = 4$ supergravity theory [14] in four dimensions coupled to six vector multiplets (from

the gravity sector) and $N = 4$ supersymmetric Yang-Mills theory. In the generalized reduction some of the six vector multiplets would provide modes for massive states of the gravity multiplet, and a rather complicated reshuffling would occur.

With these comments as motivation, consider a pure gauge theory coupled to gravity in $D + E$ dimensions. The action for the gauge fields is

$$S = -\frac{1}{4} \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} \hat{g}^{\hat{\mu}\hat{\rho}} \hat{g}^{\hat{\nu}\hat{\lambda}} \hat{G}_{\hat{\mu}\hat{\nu}}^i \hat{G}_{\hat{\rho}\hat{\lambda}}^i, \quad (71)$$

where

$$\hat{G}_{\hat{\mu}\hat{\nu}}^i = \partial_{\hat{\mu}} \hat{B}_{\hat{\nu}}^i - \partial_{\hat{\nu}} \hat{B}_{\hat{\mu}}^i + g c^i_{jk} \hat{B}_{\hat{\mu}}^j \hat{B}_{\hat{\nu}}^k, \quad (72)$$

and the c^i_{jk} are the structure constants of a gauge group H. In addition to general coordinate invariance, this theory is invariant under local gauge transformations, the infinitesimal transformations being of the form

$$\delta \hat{B}_{\hat{\mu}}^i = \partial_{\hat{\mu}} \Lambda^i - g c^i_{jk} \Lambda^j \hat{B}_{\hat{\mu}}^k. \quad (73)$$

Ordinary dimensional reduction to D dimensions gives a vector and E scalars, all massless and in the adjoint representation of H. Since the resulting potential has asymmetrical minima as well as the symmetrical ones, some of the scalars can be eaten by vectors that become massive. In fact, it was pointed out in refs. [7,15] that this Higgs effect can be reinterpreted as due to a generalized reduction in which some of the vector fields acquire mass as a consequence of a y dependence controlled by the group H.

Now let us investigate what happens to eq. (71) when the generalized reduction of sect. 2 is applied. It is necessary that the resulting theory have the invariances of the group G, gauged by the A_{μ}^{α} fields, and the group H, gauged by the B_{μ}^i fields. Following the rules for y dependence in subsect. 2.3 one considers

$$\hat{B}_{\hat{\mu}}^i(x, y) = \tilde{B}_{\mu}^i(x), \quad (74a)$$

$$\hat{B}_{\hat{\alpha}}^i(x, y) = U_{\alpha}^{\beta}(y) B_{\beta}^i(x). \quad (74b)$$

This y dependence involves the Greek indices of the group G only and not the indices of the group H. A more general y dependence involving both groups in a non-factorizable way could be considered, as in the study of instantons and magnetic monopoles. Only this simple form will be considered here, even though we recognize that a wider class of possibilities may exist.

The field $\tilde{B}_{\mu}^i(x)$ is not invariant under ξ^{α} transformations. Therefore it is useful to define

$$B_{\mu}^i(x) = \tilde{B}_{\mu}^i(x) - 2\kappa A_{\mu}^{\alpha}(x) B_{\alpha}^i(x), \quad (75)$$

which is invariant under ξ^{α} transformations. This formula is equivalent to the tangent-space relation

$$\delta^{-\gamma} V_{\tau}^{\mu}(x) B_{\mu}^i(x) = \hat{V}_{\tau}^{\hat{\mu}}(x, y) \hat{B}_{\hat{\mu}}^i(x, y). \quad (76)$$

To obtain the Yang-Mills action in D dimensions it is convenient to compute the “Latin” field strengths

$$\hat{G}_{rs}^i = \hat{V}_r^\mu \hat{V}_s^\nu \hat{G}_{\mu\nu}^i. \quad (77)$$

The y dependence cancels in this equation and one obtains

$$\hat{G}_{rs}^i = \delta^{-2\gamma} V_r^\mu V_s^\nu (G_{\mu\nu}^i + 2\kappa F_{\mu\nu}^\alpha B_\alpha^i), \quad (78)$$

where

$$G_{\mu\nu}^i = \partial_\mu B_\nu^i - \partial_\nu B_\mu^i + g c^i_{jk} B_\mu^j B_\nu^k \quad (79)$$

is the x -space field strength for the group H . $F_{\mu\nu}^\alpha$ is the field strength for the group G defined in eq. (36). It is easy to see that $\hat{G}_{rs}^i$ is invariant under G and covariant with respect to H . Similarly one finds that

$$\hat{G}_{ar}^i = -\delta^{-\gamma} \Phi_a^\alpha V_r^\nu \mathcal{D}_\nu B_\alpha^i, \quad (80)$$

$$\hat{G}_{ab}^i = -\Phi_a^\alpha \Phi_b^\beta (f^\gamma_{\alpha\beta} B_\gamma^i - g c^i_{jk} B_\alpha^j B_\beta^k), \quad (81)$$

where $\mathcal{D}_\nu$ is covariant under $G \otimes H$:

$$\mathcal{D}_\nu B_\alpha^i = \partial_\nu B_\alpha^i + g c^i_{jk} B_\alpha^j B_\nu^k - 2\kappa f^\gamma_{\alpha\beta} B_\gamma^i A_\nu^\beta. \quad (82)$$

Assembling these results, eq. (71) reduces to the D -dimensional formula

$$\begin{aligned} S = \int d^D x \quad & V \left\{ -\frac{1}{4} \delta^{2/(D-2)} (G_{\mu\nu}^i + 2\kappa F_{\mu\nu}^\alpha B_\alpha^i) (G^{\mu\nu i} + 2\kappa F^{\mu\nu\beta} B_\beta^i) \right. \\ & + \frac{1}{2} h^{\alpha\beta} g^{\mu\nu} \mathcal{D}_\mu B_\alpha^i \mathcal{D}_\nu B_\beta^i \\ & - \frac{1}{4} \delta^{-2/(D-2)} h^{\alpha\alpha'} h^{\beta\beta'} (f^\gamma_{\alpha\beta} B_\gamma^i - g c^i_{jk} B_\alpha^j B_\beta^k) \\ & \left. \times (f^{\gamma'}_{\alpha'\beta'} B_{\gamma'}^i - g c^i_{j'k'} B_{\alpha'}^{j'} B_{\beta'}^{k'}) \right\}. \end{aligned} \quad (83)$$

The potential given by the negative of the last term in eq. (83) is easily seen to be non-negative. In particular, it vanishes for the symmetrical vacuum $B_\alpha^i = 0$, which is a possible solution of the classical equations of motion. Taking this choice for the quantum mechanical vacuum state, the spectrum consists of a massless vector B_μ and E scalar fields B_α , each in the adjoint representation of H . Under the group G , B_μ is a singlet and the B_α 's belong to the adjoint representation. In addition to describing various interactions, the potential contains the mass term

$$\frac{1}{2} B_\alpha^i (\mathcal{M}^2)_{\alpha\beta} B_\beta^i = \frac{1}{4} f^{\gamma\gamma'}_{\alpha\beta} f^{\gamma'\gamma}_{\alpha\beta} B_\gamma^i B_{\gamma'}^i. \quad (84)$$

Therefore in the case where G is a flat group in an odd number of dimensions described by eq. (56), one finds that the scalar field B_1^i remains massless, while B_2^i and B_3^i describe charged scalar fields of mass m_1 , B_4^i and B_5^i describe charged scalar fields of mass m_2 , and so forth: the same pattern of masses as for the vector field A_μ^α . Asymmetric vacuum states for which $\langle B_1^i \rangle \neq 0$ can also be considered. In this case some of the components of B_1^i are eaten by the corresponding components of B_μ^i , which become massive. The important point to notice in this case is that none

of the masses in the gravity sector, i.e., corresponding to A_μ^α and $h_{\alpha\beta}$, is affected.

When this analysis is applied to the 10-dimensional supersymmetric gauge theory it is clear from the mass splittings that the resulting $N = 4$ Yang-Mills theory in 4 dimensions has broken supersymmetry. The flat space limit ($\kappa \rightarrow 0$) gives a theory without gravity, but also without invariance under supersymmetry transformations. This occurs because the coupling to supergravity is crucial for the theory to have spontaneously broken supersymmetry. This situation is possible because the global supersymmetry variation of the Goldstone spinor fields is proportional to κ^{-1} .

3.3. Second-rank antisymmetric tensor field

Ten-dimensional supergravity theory contains an antisymmetric tensor gauge field $\hat{B}_{\hat{\mu}\hat{\nu}}$. The theory is invariant under the Abelian group of gauge transformations

$$\delta_\Lambda \hat{B}_{\hat{\mu}\hat{\nu}} = \partial_{\hat{\mu}} \Lambda_{\hat{\nu}} - \partial_{\hat{\nu}} \Lambda_{\hat{\mu}} . \quad (85)$$

The gauge-invariant field strength $\hat{G}_{\hat{\mu}\hat{\nu}\hat{\rho}}$ is given by

$$\hat{G}_{\hat{\mu}\hat{\nu}\hat{\rho}} = \partial_{\hat{\mu}} \hat{B}_{\hat{\nu}\hat{\rho}} + \partial_{\hat{\nu}} \hat{B}_{\hat{\rho}\hat{\mu}} + \partial_{\hat{\rho}} \hat{B}_{\hat{\mu}\hat{\nu}} , \quad (86)$$

and its kinetic term in the presence of gravity is

$$S = \frac{1}{12} \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} \hat{g}^{\hat{\mu}\hat{\mu}'} \hat{g}^{\hat{\nu}\hat{\nu}'} \hat{g}^{\hat{\rho}\hat{\rho}'} \hat{G}_{\hat{\mu}\hat{\nu}\hat{\rho}} \hat{G}_{\hat{\mu}'\hat{\nu}'\hat{\rho}'} . \quad (87)$$

The dimensional reduction of eq. (87) gives a theory with an antisymmetric tensor field $B_{\mu\nu}$, E vector fields $B_{\mu\alpha}$, and $\frac{1}{2}E(E-1)$ scalar fields $B_{\alpha\beta}$. The two sets of vectors A_μ^α and $B_{\mu\alpha}$ both belong to the adjoint representation of the group G , which seems somewhat unusual. One may wonder whether the dimensional reduction method does not run into problems in such a case. In fact everything turns out to be consistent because there are both ξ^α and Λ_α invariances, which together are sufficient to decouple all negative metric states.

As in eq. (74), one specifies the y dependence of the $\hat{B}$ fields by setting

$$\hat{B}_{\mu\nu}(x, y) = B_{\mu\nu}(x) , \quad (88a)$$

$$\hat{B}_{\mu\alpha}(x, y) = U_\alpha^\beta(y) \tilde{B}_{\mu\beta}(x) , \quad (88b)$$

$$\hat{B}_{\alpha\beta}(x, y) = U_\alpha^\gamma(y) U_\beta^\delta(y) B_{\gamma\delta}(x) , \quad (88c)$$

and of the gauge transformations by setting

$$\Lambda_\mu(x, y) = \tilde{\Lambda}_\mu(x) , \quad (89a)$$

$$\Lambda_\alpha(x, y) = U_\alpha^\beta(y) \tilde{\Lambda}_\beta(x) . \quad (89b)$$

The gauge transformation rules are then

$$\delta_\Lambda \tilde{B}_{\mu\nu}(x) = \partial_\mu \tilde{\Lambda}_\nu - \partial_\nu \tilde{\Lambda}_\mu , \quad (90a)$$

$$\delta_\Lambda \tilde{B}_{\mu\alpha}(x) = \partial_\mu \Lambda_\alpha , \quad (90b)$$

$$\delta_\Lambda B_{\alpha\beta}(x) = -f^{\gamma}_{\alpha\beta} \Lambda_\gamma . \quad (90c)$$

However, $\tilde{B}_{\mu\nu}$ and $\tilde{B}_{\mu\alpha}$ transform under ξ^α transformations in a noncovariant fashion involving derivatives of ξ^α . Therefore in analogy with eq. (75) one defines x -dependent fields that transform covariantly under ξ^α transformations

$$B_{\mu\nu}(x) = \tilde{B}_{\mu\nu} - 2\kappa(A_\mu^\alpha \tilde{B}_{\alpha\nu} + A_\nu^\alpha \tilde{B}_{\mu\alpha}) + 4\kappa^2 A_\mu^\alpha A_\nu^\beta B_{\alpha\beta} , \quad (91a)$$

$$B_{\mu\alpha}(x) = \tilde{B}_{\mu\alpha} - 2\kappa A_\mu^\beta B_{\beta\alpha} . \quad (91b)$$

One then has the combined transformation laws

$$\delta B_{\mu\nu} = \partial_\mu \Lambda_\nu - \partial_\nu \Lambda_\mu + 2\kappa F_{\mu\nu}^\alpha \Lambda_\alpha , \quad (92a)$$

$$\delta B_{\mu\alpha} = \xi^\beta f^{\gamma}_{\alpha\beta} B_{\mu\gamma} + \mathcal{D}_\mu \Lambda_\alpha , \quad (92b)$$

$$\delta B_{\alpha\beta} = \xi^\delta (f^{\gamma}_{\alpha\delta} B_{\gamma\beta} + f^{\gamma}_{\beta\delta} B_{\alpha\gamma}) - f^{\gamma}_{\alpha\beta} \Lambda_\gamma , \quad (92c)$$

where

$$\Lambda_\mu = \tilde{\Lambda}_\mu - 2\kappa A_\mu^\alpha \Lambda_\alpha , \quad (93)$$

and

$$\mathcal{D}_\mu \Lambda_\alpha = \partial_\mu \Lambda_\alpha - 2\kappa f^{\gamma}_{\alpha\beta} A_\mu^\beta \Lambda_\gamma . \quad (94)$$

The transformation laws in eq. (92) exhibit a rather subtle intertwining of the group G (with parameters ξ^α) and the abelian gauge group (with parameters Λ_α). We note in particular that eq. (92c) gives a shift of the scalar fields $B_{\alpha\beta}$ by an amount involving the parameters of the one group and the structure constants of the other. The Λ_α and ξ^α transformations together form a $2E$ -dimensional algebra given by

$$[\delta_{\xi_1}, \delta_{\xi_2}] = \delta_{\xi_3} , \quad (95a)$$

$$[\delta_\xi, \delta_\Lambda] = \delta_{\Lambda'} , \quad (95b)$$

$$[\delta_{\Lambda_1}, \delta_{\Lambda_2}] = 0 , \quad (95c)$$

where

$$\xi_3^\gamma = f^{\gamma}_{\alpha\beta} \xi_1^\alpha \xi_2^\beta , \quad (96)$$

and

$$\Lambda'_\alpha = f^{\gamma}_{\alpha\beta} \Lambda_\gamma \xi^\beta . \quad (97)$$

This algebra has an invariant Abelian subalgebra formed by the Λ transformations and therefore it is not semisimple. In ordinary Yang-Mills theories it is not possible

to use such an algebra because the associated Cartan-Killing metric is non-invertible and cannot be used to contract internal indices of the field strengths in forming an action. However, the same remark applies here as in subsect. 2.4, namely that the internal group indices are contracted through the scalar fields $h^{\alpha\beta}$, which have their own transformation laws. Thus the scalar fields play a crucial role in ensuring the invariance of the action under the various transformation laws.

By reducing eq. (86), one is led to define the y -independent field strengths

$$G_{\alpha\beta\gamma} = -f^\delta_{\alpha\beta} B_{\delta\gamma} + \text{cyc. perms} , \quad (98a)$$

$$G_{\mu\alpha\beta} = \mathcal{D}_\mu B_{\alpha\beta} + f^\gamma_{\alpha\beta} B_{\mu\gamma} , \quad (98b)$$

$$G_{\mu\nu\alpha} = \mathcal{D}_\mu B_{\nu\alpha} - \mathcal{D}_\nu B_{\mu\alpha} + 2\kappa F^\beta_{\mu\nu} B_{\alpha\beta} , \quad (98c)$$

$$G_{\mu\nu\rho} = \partial_\mu B_{\nu\rho} + 2\kappa F^\alpha_{\nu\rho} B_{\alpha\mu} + \text{cyc. perms} , \quad (98d)$$

where the covariant derivatives in these formulas are

$$\mathcal{D}_\mu B_{\alpha\beta} = \partial_\mu B_{\alpha\beta} - 2\kappa A^\gamma_\mu [f^\delta_{\beta\gamma} B_{\alpha\delta} + f^\delta_{\alpha\gamma} B_{\delta\beta}] , \quad (99a)$$

$$\mathcal{D}_\mu B_{\nu\alpha} = \partial_\mu B_{\nu\alpha} - 2\kappa f^\delta_{\alpha\gamma} A^\gamma_\mu B_{\nu\delta} . \quad (99b)$$

It is easy to see, using eqs. (30c), (36), and (92) that these field strengths are invariant under Λ transformations and covariant tensors under ξ^α transformations. One must use the Jacobi identity for the group G to prove the invariance of $G_{\alpha\beta\gamma}$ under Λ_δ transformations.

The reduced form of eq. (87), manifestly invariant under both Λ and ξ transformations, is

$$\begin{aligned} S = \int d^D x \quad & V \left(\frac{1}{12} \delta^{4/(D-2)} G_{\mu\nu\rho} G^{\mu\nu\rho} - \frac{1}{4} \delta^{2/(D-2)} h^{\alpha\alpha'} G_{\mu\nu\alpha} G^{\mu\nu}_{\alpha'} \right. \\ & + \frac{1}{4} h^{\alpha\alpha'} h^{\beta\beta'} G_{\mu\alpha\beta} G^\mu_{\alpha'\beta'} \\ & \left. - \frac{1}{12} \delta^{-2/(D-2)} h^{\alpha\alpha'} h^{\beta\beta'} h^{\gamma\gamma'} G_{\alpha\beta\gamma} G_{\alpha'\beta'\gamma'} \right) . \end{aligned} \quad (100)$$

This formula contains mass terms of the scalar and vector fields as well as various kinetic terms and interactions. Some of the $\frac{1}{2}E(E-1)$ scalar fields $B_{\alpha\beta}$ are eaten to give mass to vector fields $B_{\mu\alpha}$. The mass matrix for the vector field $B_{\mu\alpha}$ turns out to be identical to that of the vector fields A^α_μ , and therefore the pattern of masses is also the same. In the case of a reduction from ten to four dimensions $E=6$. Since this is an even number at least two of the A^α_μ fields and two of the $B_{\mu\alpha}$ fields must be massless if the flat groups identified in subsect. 2.4 are the only ones possible. To overcome this potential embarrassment it may be necessary to find new flat groups (if there are any) or to extend the Abelian group of Λ_α transformations to a non-Abelian group (if that is possible).

3.4. Third-rank antisymmetric tensor field

In 11-dimensional supergravity theory [6], the Bose sector consists of the *vielbein* (*elfbein*) and a third-rank antisymmetric tensor field $\hat{B}_{\hat{\mu}\hat{\nu}\hat{\rho}}$. Applying ordinary dimensional reduction and duality transformations to this theory one obtains a 4-dimensional theory with exact SO(8) invariance (rather than just the SO(7) one might have expected from the extra dimensions), and all the fields are massless. In addition, the full $N = 8$ supergravity theory [4,16] is invariant under eight supersymmetry transformations, which is the largest number possible in four dimensions.

The generalized dimensional reduction method breaks the SO(8) symmetry, so in this case we know of no compelling reason to perform the duality transformations required to exhibit an SO(8) symmetric limit. Rather we perform a reduction along the lines described previously, introducing y dependence in the fields $\hat{B}_{\hat{\mu}\hat{\nu}\hat{\rho}}$ through the U matrix, just as in eqs. (74) and (88). Since the techniques are completely analogous to those used for the second rank antisymmetric tensor field in the previous section, it is sufficient to just summarize the results.

Starting from the kinetic term

$$S = -\frac{1}{48} \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} \hat{G}_{\hat{\mu}\hat{\nu}\hat{\rho}\hat{\lambda}} \hat{G}^{\hat{\mu}\hat{\nu}\hat{\rho}\hat{\lambda}}, \quad (101)$$

where

$$\hat{G}_{\hat{\mu}\hat{\nu}\hat{\rho}\hat{\lambda}} = \partial_{\hat{\mu}} \hat{B}_{\hat{\nu}\hat{\rho}\hat{\lambda}} - \partial_{\hat{\lambda}} \hat{B}_{\hat{\mu}\hat{\nu}\hat{\rho}} + \partial_{\hat{\rho}} \hat{B}_{\hat{\lambda}\hat{\mu}\hat{\nu}} - \partial_{\hat{\nu}} \hat{B}_{\hat{\rho}\hat{\lambda}\hat{\mu}}, \quad (102)$$

and indices are raised with the inverse metric $\hat{g}^{\hat{\mu}\hat{\nu}}$, one obtains the reduced action

$$\begin{aligned} S = \int d^D x \, V \{ & -\frac{1}{48} \delta^{6/(D-2)} G_{\mu\nu\rho\lambda} G^{\mu\nu\rho\lambda} + \frac{1}{12} \delta^{4/(D-2)} G_{\mu\nu\rho\alpha} G^{\mu\nu\rho\alpha} \\ & - \frac{1}{8} \delta^{2/(D-2)} G_{\mu\nu\alpha\beta} G^{\mu\nu\alpha\beta} + \frac{1}{12} G_{\mu\alpha\beta\gamma} G^{\mu\alpha\beta\gamma} \\ & - \frac{1}{48} \delta^{-2/(D-2)} G_{\alpha\beta\gamma\delta} G^{\alpha\beta\gamma\delta} \}, \end{aligned} \quad (103)$$

where the field strengths, covariant with respect to ξ^α transformations, are

$$G_{\alpha\beta\gamma\delta} = -f^\epsilon{}_{\alpha\beta} B_{\epsilon\gamma\delta} + 5 \text{ perms}, \quad (104a)$$

$$G_{\mu\alpha\beta\gamma} = \mathcal{D}_\mu B_{\alpha\beta\gamma} + (f^\delta{}_{\alpha\beta} B_{\mu\delta\gamma} + 2 \text{ perms}), \quad (104b)$$

$$G_{\mu\nu\alpha\beta} = \mathcal{D}_\mu B_{\nu\alpha\beta} - \mathcal{D}_\nu B_{\mu\alpha\beta} + 2\kappa F^\gamma_{\mu\nu} B_{\alpha\beta\gamma} - f^\gamma{}_{\alpha\beta} B_{\mu\nu\gamma}, \quad (104c)$$

$$G_{\mu\nu\rho\alpha} = \mathcal{D}_\mu B_{\nu\rho\alpha} + 2\kappa F^\beta_{\nu\rho} B_{\mu\alpha\beta} + 2 \text{ perms}, \quad (104d)$$

$$G_{\mu\nu\rho\lambda} = (\partial_\mu B_{\nu\rho\lambda} + 3 \text{ perms}) + (2\kappa F^\alpha_{\mu\nu} B_{\rho\lambda\alpha} + 5 \text{ perms}). \quad (104e)$$

The B fields in eq. (104) have been constructed as linear combinations of the original x -dependent fields $\hat{B}_{\hat{\mu}\hat{\nu}\hat{\rho}}(x)$ that appear in the reduction by formulas analogous

to eq. (91):

$$B_{\mu\alpha\beta} = \tilde{B}_{\mu\alpha\beta} - 2\kappa A_\mu^\gamma \tilde{B}_{\gamma\alpha\beta} , \quad (105a)$$

$$B_{\mu\nu\alpha} = \tilde{B}_{\mu\nu\alpha} - 2\kappa A_\mu^\beta \tilde{B}_{\beta\nu\alpha} - 2\kappa A_\nu^\beta \tilde{B}_{\mu\beta\alpha} + 4\kappa^2 A_\mu^\beta A_\nu^\gamma \tilde{B}_{\beta\gamma\alpha} , \quad (105b)$$

$$B_{\mu\nu\rho} = \tilde{B}_{\mu\nu\rho} + (-2\kappa A_\mu^\alpha \tilde{B}_{\alpha\nu\rho} + 4\kappa^2 A_\mu^\alpha A_\nu^\beta \tilde{B}_{\alpha\beta\rho} + 2 \text{ perms}) - 8\kappa^3 A_\mu^\alpha A_\nu^\beta A_\rho^\gamma \tilde{B}_{\alpha\beta\gamma} . \quad (105c)$$

The Abelian gauge invariance of eq. (102) under the transformation

$$\delta \tilde{B}_{\hat{\mu}\hat{\nu}\hat{\rho}} = \partial_{\hat{\mu}} \Lambda_{\hat{\nu}\hat{\rho}} + \partial_{\hat{\rho}} \Lambda_{\hat{\mu}\hat{\nu}} + \partial_{\hat{\nu}} \Lambda_{\hat{\rho}\hat{\mu}} , \quad (106)$$

results in the transformation formulas

$$\delta B_{\alpha\beta\gamma} = -f^\delta{}_{\alpha\beta} \Lambda_{\delta\gamma} + 2 \text{ perms} , \quad (107a)$$

$$\delta B_{\mu\alpha\beta} = \mathcal{D}_\mu \Lambda_{\alpha\beta} + f^\gamma{}_{\alpha\beta} \Lambda_{\mu\gamma} , \quad (107b)$$

$$\delta B_{\mu\nu\alpha} = \mathcal{D}_\mu \Lambda_{\nu\alpha} - \mathcal{D}_\nu \Lambda_{\mu\alpha} + 2\kappa F_{\mu\nu}^\beta \Lambda_{\beta\alpha} , \quad (107c)$$

$$\delta B_{\mu\nu\rho} = \partial_\mu \Lambda_{\nu\rho} + 2\kappa F_{\mu\nu}^\alpha \Lambda_{\alpha\rho} + 2 \text{ perms} . \quad (107d)$$

The field strengths in eq. (104) are invariant under these transformations. Altogether, as before, there are enough invariances to decouple all negative metric states. The various masses can be read off as before.

One interesting question, still to be explored, is whether the $\frac{1}{2}E(E+1)$ dimensional algebra generated by the ξ^α and $\Lambda_{\alpha\beta}$ transformations can be extended to a richer structure than described here. We know that dual transformations must be performed before they can be combined to describe the group $O(8)$. It is also quite clear that the group $O(8)$ is one that should *not* be gauged since it would give rise to a potential unbounded from below.

3.5. Reduction of a spin- $\frac{3}{2}$ field

In the 11-dimensional supergravity theory the entire fermionic sector consists of a Majorana 32-component spin- $\frac{3}{2}$ field $\psi_{\hat{\mu}}(x, y)$. Ordinary dimensional reduction to four dimensions gives 8 Majorana spin- $\frac{3}{2}$ fields and 56 Majorana spin- $\frac{1}{2}$ fields, all massless. With generalized dimensional reduction the situation changes since 8 of the spin- $\frac{1}{2}$ fields are eaten by the 8 gravitinos, which become massive.

The kinetic term for the 11-dimensional spin- $\frac{3}{2}$ field is

$$S = -\frac{1}{2}i \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} \bar{\psi}_{\hat{m}} \Gamma^{\hat{m}\hat{n}\hat{p}} \mathcal{D}_{\hat{n}} \psi_{\hat{p}} , \quad (108)$$

where

$$\mathcal{D}_{\hat{n}} \psi_{\hat{p}} = (\hat{V}_{\hat{n}}^\mu \partial_{\hat{\mu}} + \frac{1}{4} \hat{\omega}_{\hat{n}\hat{r}\hat{s}} \Gamma^{\hat{r}\hat{s}}) \psi_{\hat{p}} + \hat{\omega}_{\hat{n}\hat{p}}^{\hat{q}} \psi_{\hat{q}} . \quad (109)$$

As in ref. [6], the Dirac matrices are 32×32 matrices satisfying

$$\{\Gamma^{\hat{r}}, \Gamma^{\hat{s}}\} = 2\eta^{\hat{r}\hat{s}}, \quad (110)$$

$$\Gamma^{\hat{r}\hat{s}} \dots \hat{z} = \begin{cases} \Gamma^{\hat{r}} \Gamma^{\hat{s}} \dots \Gamma^{\hat{z}}, & \text{all indices different,} \\ 0, & \text{otherwise.} \end{cases} \quad (111)$$

Note that flat Latin indices are used in eqs. (108) and (109). Since $\psi_{\hat{p}}$ bears no Greek indices the y dependence is trivial

$$\psi_{\hat{p}}(x, y) = \psi_{\hat{p}}(x). \quad (112)$$

The interesting fact is that masses are generated for these fields even though they are y independent because the fully internal component of the connection, $\omega_{c,ab}$ is non-zero in the weak field limit. Using eq. (35f) one finds in this limit the constant value

$$\omega_{c,ab} = \frac{1}{2}(f^c_{ab} + f^b_{ac} - f^a_{bc}). \quad (113)$$

In the case of a flat group given by a 6×6 matrix M of the form in eq. (56) with three mass parameters,

$$\omega_{c,ab} = \delta_c^1 M_a^b. \quad (114)$$

Using this formula one notes that the supersymmetry variation of the ψ_c fields contains a constant term

$$\delta\psi_c(x) = \frac{1}{4\kappa} \delta_{c1} M_a^b \Gamma^{ab} \epsilon(x) + \dots \quad (115)$$

When $M\Gamma$ is non singular the 8 spin- $\frac{1}{2}$ fields $\psi_1(x)$ are Goldstone fermions, which can be set to zero by choice of an appropriate supersymmetry gauge [★].

Having noted the presence of 8 Goldstone fermions in the reduced theory, one can investigate the mass pattern of the remaining fields. For this purpose it is convenient to relabel the seven y coordinates as follows

$$y^1 = y, \quad (116a)$$

$$y^2, y^4, y^6 = w^1, w^2, w^3, \quad (116b)$$

$$y^3, y^5, y^7 = z^1, z^2, z^3, \quad (116c)$$

and to choose the representation

$$\Gamma^\mu = \gamma^\mu \otimes 1_2 \otimes 1_4, \quad (117a)$$

$$\Gamma^y = \gamma^5 \otimes i\tau_2 \otimes 1_4, \quad (117b)$$

$$\Gamma^{wi} = \gamma^5 \otimes \tau_1 \otimes \alpha^i, \quad (117c)$$

$$\Gamma^{zi} = \gamma^5 \otimes \tau_3 \otimes \beta^i, \quad (117d)$$

[★] This shows that in contrast with examples studied in the past, the Goldstone fermions in the $N = 8$ theory do not need to be added explicitly as Volkov-Akulov fermions [17], or as a part of a matter multiplet [18], which does not exist in the case of the $N > 4$ theories. The remarkable feature is that the modified $N = 8$ theory contains its own set of Goldstone fermions.

where α^i, β^i are six 4×4 real antisymmetric matrices satisfying the algebra

$$\{\alpha^i, \alpha^j\} = \{\beta^i, \beta^j\} = -2\delta^{ij}, \quad (118a)$$

$$[\alpha^i, \beta^j] = 0. \quad (118b)$$

An explicit representation of the α 's and β 's is given in ref. [12]. This representation has the useful property that $\alpha^i \beta^i$, for $i = 1, 2, 3$, is a diagonal matrix with entries ± 1 . The term that determines the masses of the 4-dimensional spin- $\frac{3}{2}$ fields obtained using eqs. (108), (111), and (114) is

$$\mathcal{L}_{\text{mass}} = \frac{1}{4} i \bar{\psi}_m \sigma^{mp} \Gamma^1 M_{ab} \Gamma^{ab} \psi_p. \quad (119)$$

Using the explicit form of the M matrix and eq. (117) this simplifies to

$$\mathcal{L}_{\text{mass}} = -i V \bar{\psi}_m \sigma^{mp} \gamma_5 (1_2 \otimes \mathcal{M}) \psi_p, \quad (120)$$

where

$$\mathcal{M} = \frac{1}{2} (m_1 \alpha^1 \beta^1 + m_2 \alpha^2 \beta^2 + m_3 \alpha^3 \beta^3). \quad (121)$$

The $i\gamma_5$ factor can be removed by a chiral field redefinition. In the 2×2 space spanned by the τ matrices eq. (120) has a unit matrix. Thus the 8 Majorana 4-component spin- $\frac{3}{2}$ fields $\psi_m^{(i,k)}$ ($i = 1, 2; k = 1, 2, 3, 4$) are 2-fold degenerate and group together into four Dirac spinors whose masses are given by

$$\mathcal{M}_{3/2}^k = \frac{1}{2} |m_1 \pm m_2 \pm m_3|. \quad (122)$$

The three mass parameters can be chosen such that no gravitino is massless, and $N = 8$ supersymmetry is completely broken.

Table 2

Mass spectrum of the $N = 8$ model after spontaneous symmetry breaking; $M^2 = m_1^2 + m_2^2 + m_3^2$

J	M	Number of real fields	Degeneracy	$\Sigma(2J+1)(-1)^{2J} m_J^2$
2	0	1	1	0
$\frac{3}{2}$	$\frac{1}{2} m_1 \pm m_2 \pm m_3 $	8	2	$-8M^2$
1	0	4	4	0
	m_i	12	4	$12M^2$
	$ m_i \pm m_j $	12	2	$24M^2$
$\frac{1}{2}$	$\frac{1}{2} m_1 \pm m_2 \pm m_3 \pm m_i$	48	2	$-56M^2$
0	0	8	8	0
	m_i	12	4	$4M^2$
	$2m_i$	6	2	$8M^2$
	$ m_i \pm m_j $	12	2	$8M^2$
	$ m_1 \pm m_2 \pm m_3 $	8	2	$8M^2$

The masses of the remaining 48 Majorana spin- $\frac{1}{2}$ fields are more complicated to derive, because a spin- $\frac{3}{2}$ – spin- $\frac{1}{2}$ diagonalization is required, among other things. However, based on the pattern that seems to be emerging, one might guess that the 48 Majorana spinors group into 24 Dirac spinors, with masses taking the 24 values given by

$$\mathcal{M}_{1/2} = |\mathcal{M}_{3/2}^k \pm m_i|, \quad k = 1, 2, 3, 4; i = 1, 2, 3. \quad (123)$$

The final mass spectrum of the reduced $D = 4, N = 8$ theory is exhibited in table 2. It satisfies a mass formula valid for a general spontaneously broken supersymmetry theory without a cosmological constant [19]:

$$\sum_J (-1)^{2J} (2J + 1) m_J^2 = 0. \quad (124)$$

In this sum each real field is counted with weight one. In our case, bosons and fermions each contribute $64(m_1^2 + m_2^2 + m_3^2)$.

4. Conclusion

The generalized method of dimensional reduction described here is applicable to supersymmetric theories, as has been noted on a number of occasions in the text, and provides examples of how the mass degeneracy of the extended supergravity theories ($N = 4, 8$) can be spontaneously broken without introducing a cosmological term or badly behaved potential. In these examples all the gravitinos can become massive so that all the supersymmetries are broken. Furthermore, most of the other fields also acquire mass, while the graviton necessarily remains massless.

As a mathematically consistent theory, one may feel optimistic about broken $N = 8$ supergravity theory. The symmetry breaking described here should not adversely affect the ultraviolet behavior of the theory *. Of course, there is still a long way to go to construct a physically realistic theory. Nevertheless we find it tantalizing that there are 24 Dirac spin- $\frac{1}{2}$ fields, just enough to describe six quarks each coming in three colors and six leptons (including three unobserved right-handed neutrinos). On the other hand we still tend to feel that broken $N = 4$ theory, obtained from the reduction of 10-dimensional supergravity theory, may be more promising, both because it allows the freedom of introducing a Yang-Mills group and because of its intimate relation to the spinning string theory [21]. Clearly, additional tricks are required if either of these theories is to describe nature. Still, we feel a certain satisfaction at this stage since it had been suspected that $N = 4$ and $N = 8$ supersymmetry could not be broken at all [22].

Among the many remaining problems and unanswered questions the following seem to deserve attention: (1) Are there other flat groups than those described in subsect. 2.4? (2) Are there additional mechanisms that can be employed that would

* For a discussion of the ultraviolet behavior of the massless $N = 8$ theory, see ref. [20].

lead to a gauge group containing at least $SU(3)_C \otimes U(1)_{EM}$? For example, is it possible to identify the extra dimensions as a coset space rather than as a group [23]? (3) The “photon” that has been described couples to a conserved charge that is proportional to mass. How can we get rid of this particle and obtain another massless vector that behaves like a decent photon? (4) How should parity violation and the broken $SU(2) \otimes U(1)$ of the electroweak interactions be introduced? One can only hope that the patient study of these problems will lead to further progress and that a realistic supersymmetry theory can eventually be constructed.

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